



20 EPA 608 Terms Every HVAC Tech Should Know

With Definitions, Real Examples & Exam Tips



*Free practice tests for HVAC technicians.
Study smarter, pass faster.*

EPA608PracticeTest.net

TERM 01 | Refrigerant

TERM 01

Refrigerant

A chemical used to absorb and transfer heat in a refrigeration system. Refrigerants are the working fluids found in air conditioners, refrigerators, freezers, and heat pumps.

■ **Exam Tip:** Know the common examples: R-410A, R-22, R-134a, R-1234yf.

REFRIGERANT

A chemical used to absorb and transfer heat in a refrigeration system.

EXAMPLES:

- R-410A
- R-22
- R-134a
- R-1234yf



Refrigerants are the working fluids found in air conditioners, refrigerators, freezers, and heat pumps.



TERM 02 | ODP — Ozone Depletion Potential

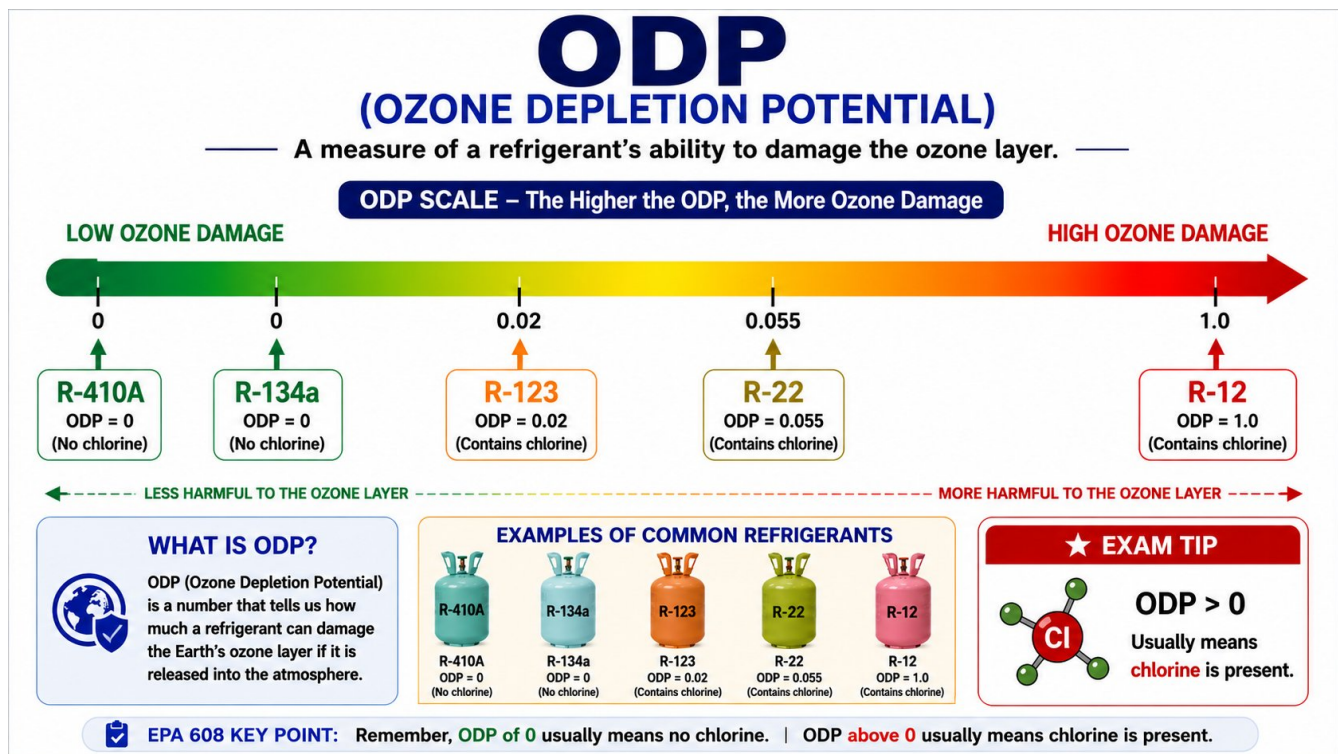
TERM 02

ODP — Ozone Depletion Potential

Measures a refrigerant's ability to damage the ozone layer. The higher the ODP, the more ozone damage it can cause. R-11 and R-12 (CFCs) have the highest ODP. R-410A and R-134a have ODP = 0.

■ **Exam Tip:** ODP above zero usually means chlorine is present in the refrigerant.

■ **Memory Trick:** ODP = Ozone Damage Potential. If there's chlorine, there's ODP.



TERM 03 | GWP — Global Warming Potential

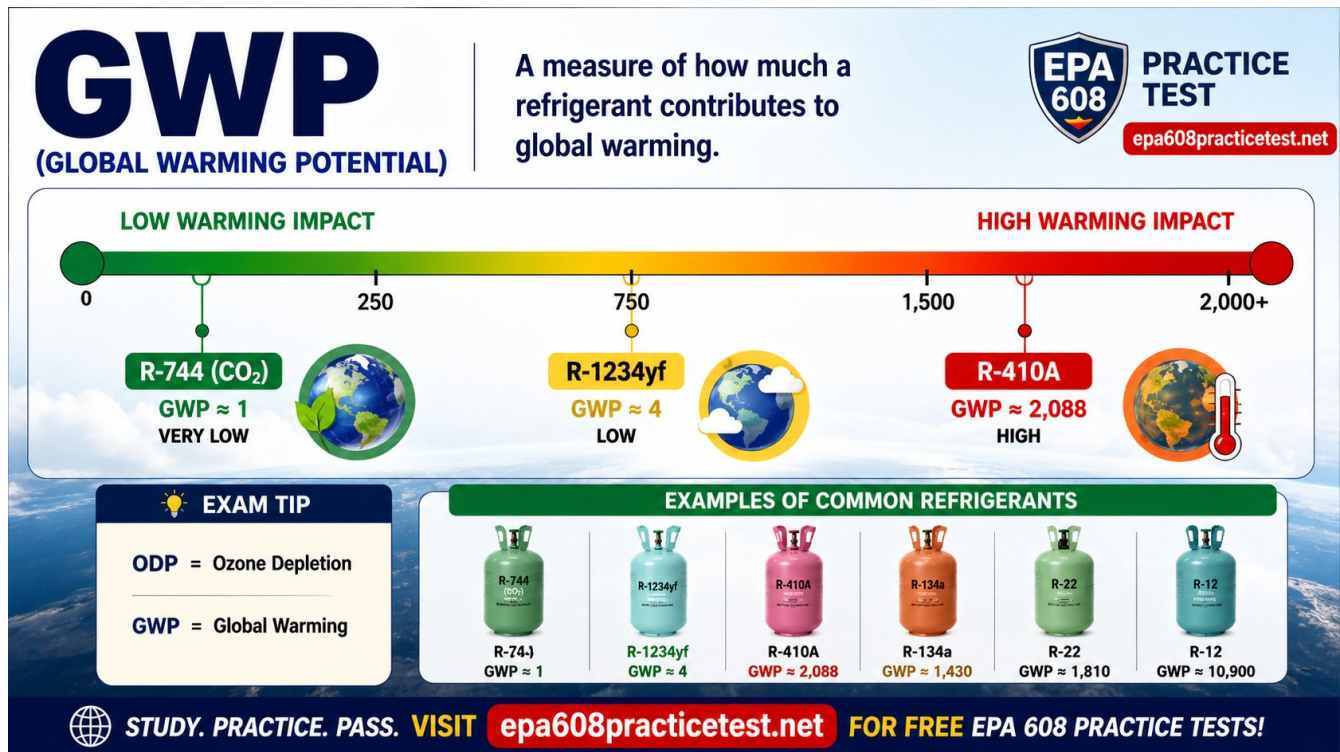
TERM 03

GWP — Global Warming Potential

Measures how much a refrigerant contributes to global warming compared to CO₂. The higher the GWP, the greater its environmental impact. R-410A has GWP $\approx$ 2,088. R-1234yf (HFO) has GWP $\approx$ 4.

■ **Exam Tip:** Higher ODP = more ozone damage. Higher GWP = more global warming impact. These are two separate measures.

■ **Memory Trick:** GWP = Global Warming Power. Higher number = worse for climate.



TERMS 04–07 | CFC • HCFC • HFC • HFO — Refrigerant Families

Refrigerants are grouped into four chemical families. Knowing the names, key properties, and examples of each family is essential for the EPA 608 exam.

TERM 04 — CFC (Chlorofluorocarbons)

Contains: Chlorine + Fluorine + Carbon

ODP: > 0 | GWP: High

Examples: R-11, R-12

Found in older refrigerators.

TERM 05 — HCFC (Hydrochlorofluorocarbons)

Contains: H + Chlorine + Fluorine + Carbon

ODP: > 0 | GWP: High

Example: R-22

Common in older AC systems.

TERM 06 — HFC (Hydrofluorocarbons)

Contains: H + Fluorine + Carbon — NO Chlorine

ODP: 0 | GWP: High

Examples: R-410A, R-134a, R-404A

Common in residential split systems.

TERM 07 — HFO (Hydrofluoroolefins)

Contains: H + Fluorine + Olefin — NO Chlorine

ODP: 0 | GWP: Very Low

Examples: R-1234yf, R-1234ze

Newest generation — found in modern cars.



REFRIGERANT FAMILIES

Know the Names. See the Examples.

**EPA 608
PRACTICE TEST**

epa608practicetest.net

CFC	HCFC	HFC	HFO
Chlorofluorocarbons Chloro - Fluoro - Carbons <u>Cl</u> <u>F</u> <u>C</u> EXAMPLES   REAL-WORLD EXAMPLE  Older refrigerators often used R-12.	Hydrochlorofluorocarbons Hydro - Chloro - Fluoro - Carbons <u>H</u> <u>Cl</u> <u>F</u> <u>C</u> EXAMPLE  REAL-WORLD EXAMPLE  Many older residential air conditioners use R-22.	Hydrofluorocarbons Hydro - Fluoro - Carbons <u>H</u> <u>F</u> <u>C</u> EXAMPLES    REAL-WORLD EXAMPLE  Many residential split systems use R-410A.	Hydrofluoroolefins Hydro - Fluoro - Olefins <u>H</u> <u>F</u> <u>O</u> EXAMPLES   REAL-WORLD EXAMPLE  Many newer cars use R-1234yf.

 **QUICK RECAP:** CFC (Chloro) – HCFC (Hydro + Chloro) – HFC (No Chloro) – HFO (New generation)

■ Quick Recap: CFC (Chloro) — HCFC (Hydro + Chloro) — HFC (No Chloro) — HFO (New generation)

TERMS 08–10 | Recovery • Recycling • Reclamation

There are three methods for handling used refrigerant. Each is a distinct process — and all three appear on the EPA 608 exam.

TERM 08

Recovery

Remove refrigerant from a system and store it in an EPA-approved recovery cylinder. No cleaning or processing takes place — just remove and store.

■ **Exam Tip:** Recovery must be performed before opening any refrigeration system for service.

■ **Memory Trick:** Recovery = REMOVE & STORE.

TERM 09

Recycling

Clean recovered refrigerant through filtration and oil separation so it can be reused in the same system or another system owned by the same owner. Recycling happens on-site and does not restore refrigerant to virgin-grade purity.

■ **Exam Tip:** Recycling does NOT meet AHRI 700 standards — only reclamation does.

■ **Memory Trick:** Recycling = CLEAN & REUSE.

TERM 10

Reclamation

Process used refrigerant at a certified reclamation facility to restore it to virgin-grade specifications as defined by AHRI Standard 700. This is the highest level of refrigerant processing.

■ **Exam Tip:** Only EPA-certified reclamation facilities can reclaim refrigerant.

■ **Memory Trick:** Reclamation = RESTORE TO LIKE-NEW.

RECOVERY → RECYCLING → RECLAMATION

THREE WAYS TO HANDLE USED REFRIGERANT



EPA 608
PRACTICE TEST
EPA608PracticeTest.net



MEMORY TRICK:

REMOVE



CLEAN



LIKE NEW

TERMS 11–12 | Evacuation • Vacuum Pump

Before adding refrigerant to any system, technicians must perform evacuation to remove air and moisture. These two terms are closely related but describe different things.

TERM 11

Evacuation

The process of removing air and moisture from a refrigeration system using a vacuum pump. The goal is a clean, dry system that is ready to receive refrigerant. Air and moisture left in the system cause pressure problems and corrosion.

■ **Exam Tip:** *Evacuation removes non-condensables (air) and moisture — both harm system performance.*

■ **Memory Trick:** Evacuation = THE PROCESS.

TERM 12

Vacuum Pump

The tool used to perform evacuation. A vacuum pump creates a deep vacuum inside the refrigeration system, which causes moisture to boil off and be removed along with air.

■ **Exam Tip:** *A vacuum pump is required equipment for any EPA 608 certified technician.*

■ **Memory Trick:** Vacuum Pump = THE TOOL.

EVACUATION + VACUUM PUMP

REMOVE AIR & MOISTURE FROM A SYSTEM

EVACUATION

THE PROCESS

Removing air and moisture from a system.



AIR

✗

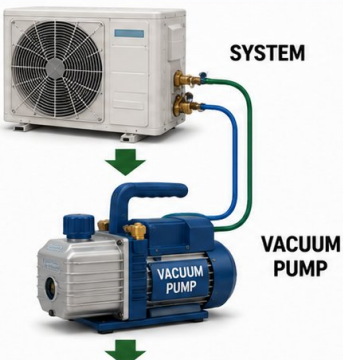


MOISTURE

✗



GOAL:
A clean, dry system
ready for refrigerant.



SYSTEM

VACUUM PUMP



**AIR & MOISTURE
REMOVED**

VACUUM PUMP

THE TOOL

Used to evacuate a refrigeration system.



 **MEMORY TRICK:** **EVACUATION** = **PROCESS** | **VACUUM PUMP** = **TOOL**

TERMS 13–15 | Compressor Burnout • Oil Sample • Major Repair

These three terms frequently appear together on the EPA 608 exam because they describe a connected chain of events in refrigeration system failures.

TERM 13

Compressor Burnout

A serious motor failure inside the compressor that contaminates the entire system with acid, carbon, and debris. One of the most damaging failures in HVAC.

■ **Exam Tip:** After a burnout, system oil and refrigerant are likely contaminated with acid.

TERM 14

Oil Sample

When a burnout is suspected, technicians take an oil sample from the system and test it for acid contamination. A positive acid test confirms that a burnout has occurred.

■ **Exam Tip:** An oil sample is used to **CONFIRM** whether a compressor burnout has taken place.

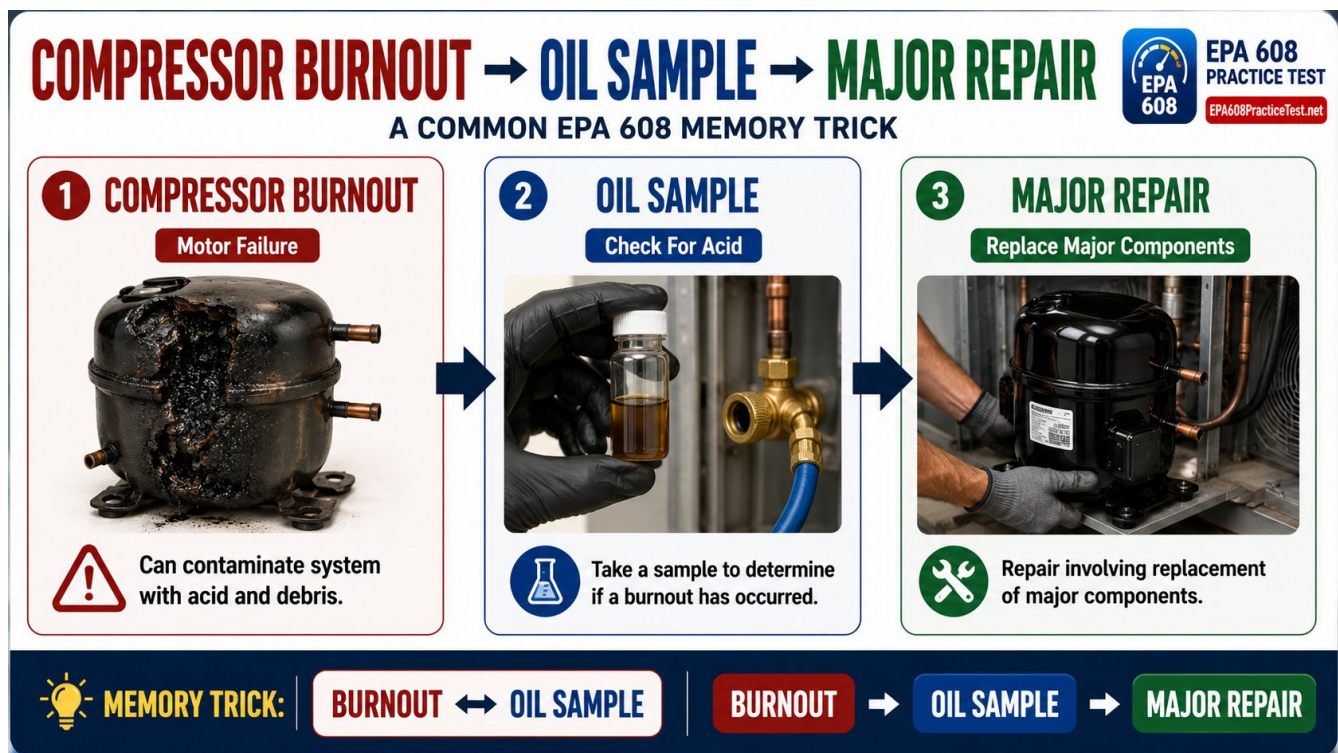
TERM 15

Major Repair

Any repair involving the replacement of major system components such as the compressor, condenser, or evaporator. A compressor burnout typically triggers a major repair.

■ **Exam Tip:** Major repairs often require complete refrigerant recovery before work begins.

■ **Memory Trick:** Burnout → Oil Sample (check for acid) → Major Repair (replace components).



TERMS 16–18 | Low-Loss Fittings • Leak Detection • Leak Repair Threshold

EPA 608 regulations place strict requirements on refrigerant leak management. These three terms cover the key concepts.

TERM 16

Low-Loss Fittings

Special service fittings that prevent refrigerant from escaping when hoses are connected to or disconnected from a system. They minimize refrigerant emissions during routine service.

■ **Exam Tip:** Required when servicing systems containing more than 5 lbs of refrigerant.

TERM 17

Leak Detection

The process of locating refrigerant leaks in a system using electronic leak detectors, UV dye, bubble solution, or ultrasonic tools. Regular leak checks are required by EPA for larger commercial systems.

■ **Exam Tip:** The detection method used must be appropriate for the refrigerant type.

TERM 18

Leak Repair Threshold

The leak rate at which EPA regulations require the owner/operator to repair leaks on certain equipment. Thresholds differ by equipment type: 10% for comfort cooling, 20% for commercial refrigeration, 30% for IPR equipment.

■ **Exam Tip:** Leak rates are expressed as % of the system's total refrigerant charge per year.

■ **Memory Trick:** Minimize Emissions → Detect Leaks → Repair Leaks.



TERM 19 | High-Pressure Refrigerants

TERM 19

High-Pressure Refrigerants

Refrigerants that operate at higher pressures within a refrigeration system. These are the most common refrigerants encountered in the field and on the EPA 608 exam. You'll find them in residential and commercial air conditioning systems.

■ **Exam Tip:** R-410A and R-22 are the two most common high-pressure refrigerants. They are covered under EPA 608 Type II certification.

■ **Memory Trick:** R-410A and R-22 = high-pressure = Type II systems.

HIGH-PRESSURE REFRIGERANTS

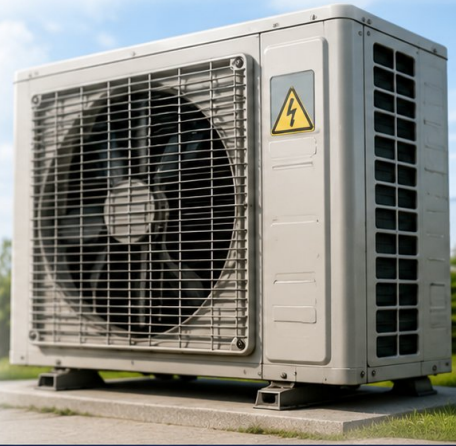
Operate at Higher Pressures



EPA 608
PRACTICE TEST

EPA608PracticeTest.net

EXAMPLES



REFRIGERANTS
THAT OPERATE AT
HIGHER PRESSURES.

R-410A • R-22



EPA 608
TYPE II
COMMON EXAMPLES

TERM 20 | Appliance

TERM 20

Appliance

Under EPA 608, an appliance is any device that contains and uses a refrigerant in its normal operation. This includes household refrigerators, freezers, heat pumps, air conditioners, and large commercial supermarket rack systems.

■ **Exam Tip:** *If it contains and uses refrigerant — it is considered an appliance under EPA 608.*

■ **Memory Trick:** Any device using refrigerant = Appliance.

APPLIANCE

Any Device That Contains and Uses Refrigerant

EPA 608 PRACTICE TEST
EPA608PracticeTest.net

EXAMPLES OF APPLIANCES



REFRIGERATOR



FREEZER



HEAT PUMP



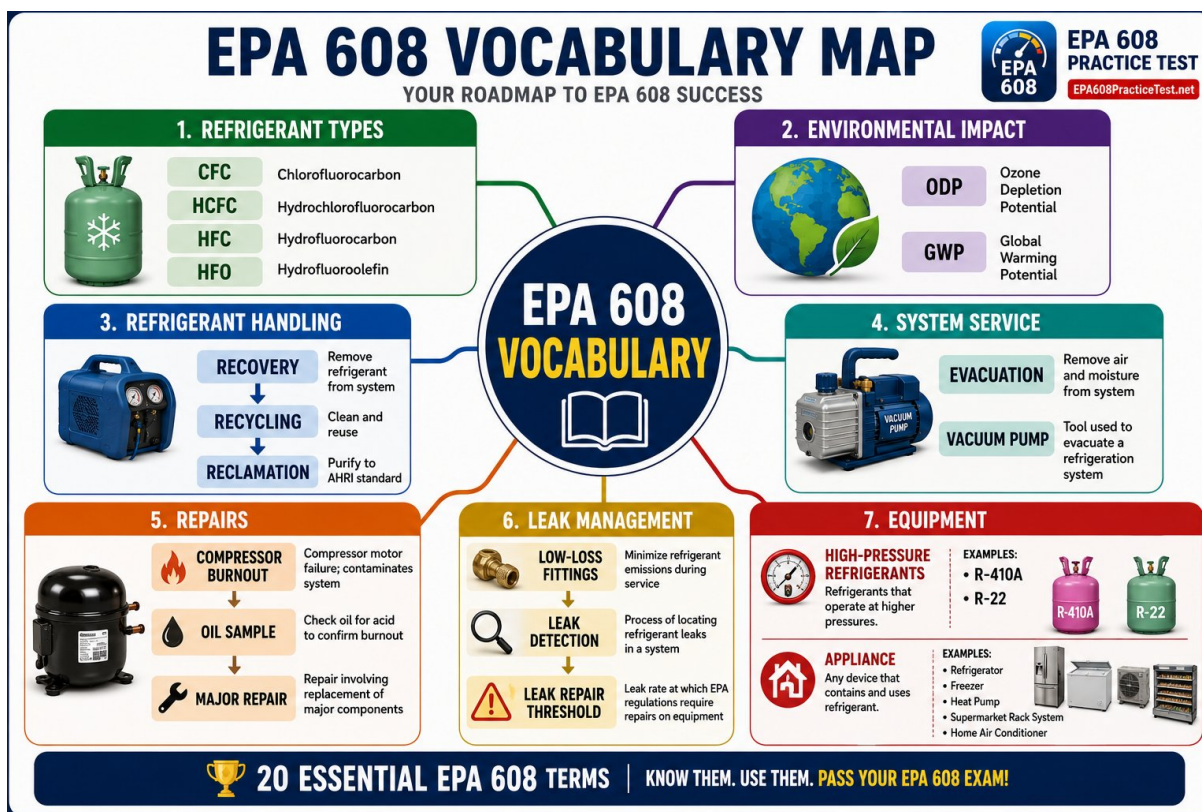
SUPERMARKET RACK SYSTEM



HOME AIR CONDITIONER

✓ ALL OF THESE ARE CONSIDERED **APPLIANCES** UNDER EPA 608.

VOCABULARY MAP | All 20 Terms at a Glance



3 Things to Remember

- ODP = Ozone Damage Potential
- GWP = Global Warming Impact
- Recovery → Recycling → Reclamation
- Compressor Burnout → Oil Sample → Major Repair

Want more free EPA 608 practice tests and study resources?

Visit: EPA608PracticeTest.net

Free practice tests • Study guides • Flashcards • Timed exams